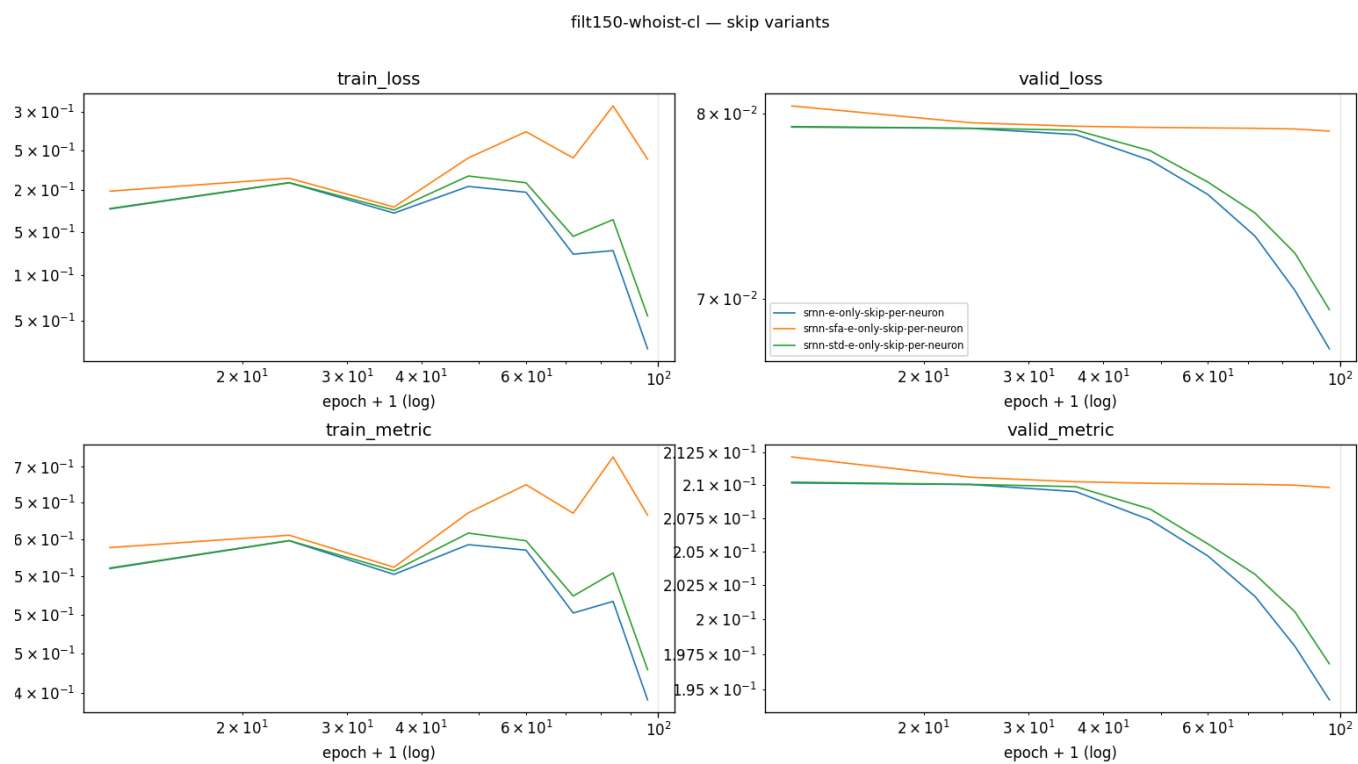


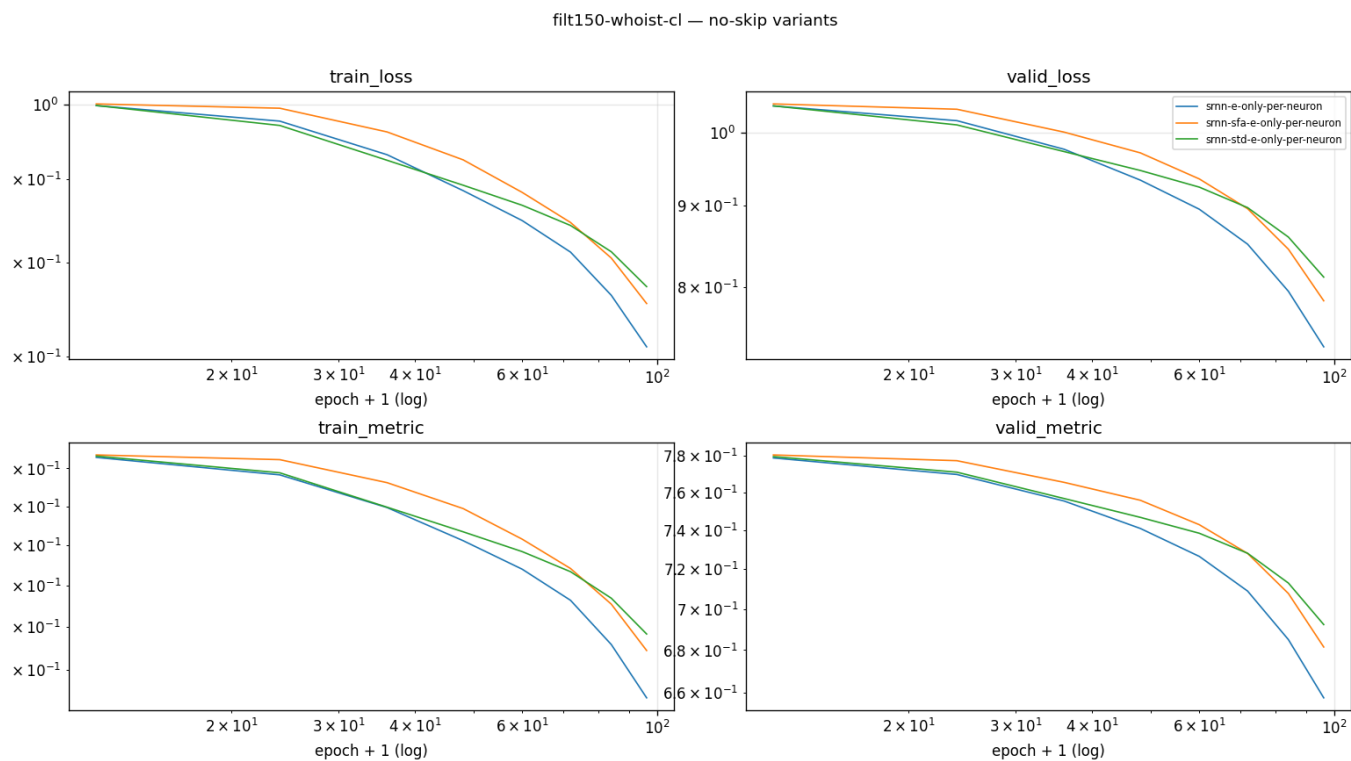
Run: flt150-whoist-cl

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07877 → nan
 srnn-sfa-e-only-skip-per-neuron: 0.09420 → nan
 srnn-std-e-only-skip-per-neuron: 0.07907 → nan

Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.98139 - nan
 srnn-sfa-e-only-per-neuron: 0.91630 - nan
 srnn-std-e-only-per-neuron: 0.98171 - nan

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

filt150-whoist-cl — effective time constants (srnn-e-only-per-neuron)

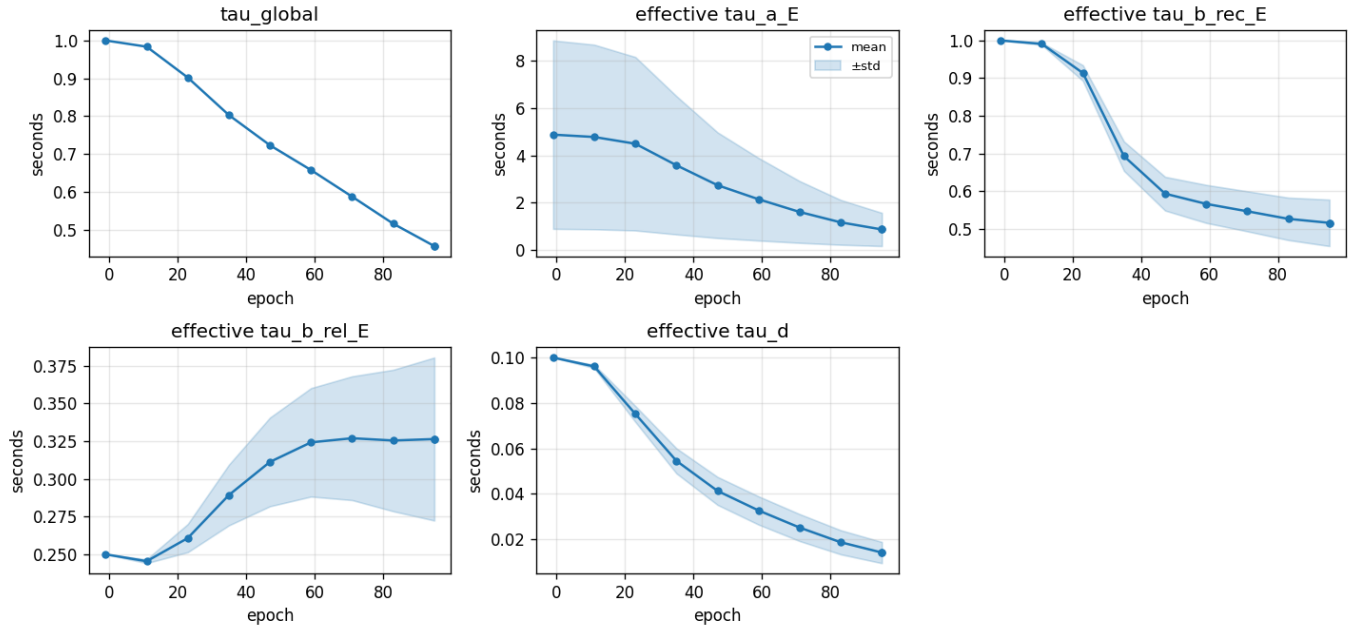
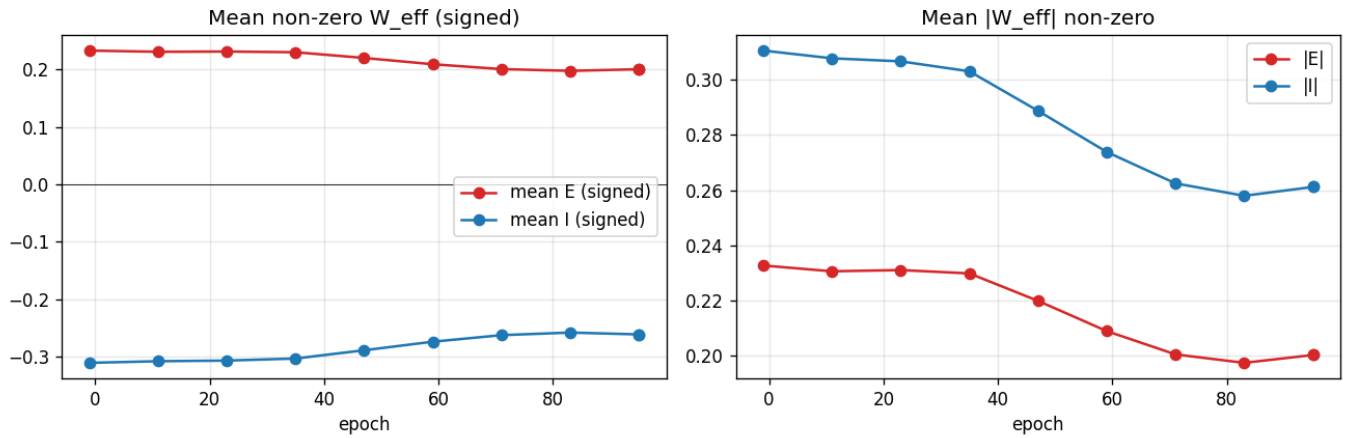


Figure 3: Tau evolution

filt150-whoist-cl — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.45741	-	-	scalar
tau_d (s)	+0.1	+0.014046	0	0.004683	300
tau_a_E[j=0] (s)	+0.25	+0.034603	0	0.008747	150
tau_a_E[j=1] (s)	+4.3832	+0.78163	0	0.03361	150
tau_a_E[j=2] (s)	+10	+1.7562	0	0.02638	150
tau_b_rec_E (s)	+1	+0.51562	0	0.0617	150
tau_b_rel_E (s)	+0.25	+0.32634	0	0.05398	150
W_eff (E src, signed)	+0.23267	+0.2002	0.07641	0.07165	15048
W_eff (I src, signed)	-0.31062	-0.26116	0.07724	0.0718	14977
W_eff (E src)	+0.23267	+0.2002	0.07641	0.07165	15048
W_eff (I src)	+0.31062	+0.26116	0.07724	0.0718	14977
W_in_eff (input neurons)	-0.0003386	+0.00077285	0.1012	0.08464	6675
c_E[j=0] (SFA coupling)	+0.05	+0.10078	0	0.02885	150
c_E[j=1] (SFA coupling)	+0.05	+0.088987	0	0.01577	150
c_E[j=2] (SFA coupling)	+0.05	+0.090157	0	0.01536	150
c_0_E[j=0] (SFA offset)	+0	-0.55652	0	0.1215	150
c_0_E[j=1] (SFA offset)	+0	-0.6329	0	0.1548	150
c_0_E[j=2] (SFA offset)	+0	-0.64094	0	0.1601	150
a_0 (threshold)	+0.35	-0.23773	0	0.1249	300
readout_weight	-2.8854e-05	-0.0045077	0.06604	0.2072	6675
readout_bias	+0	+0.0068269	0	0.1851	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

filt150-whoist-cl — effective time constants (srnn-e-only-skip-per-neuron)

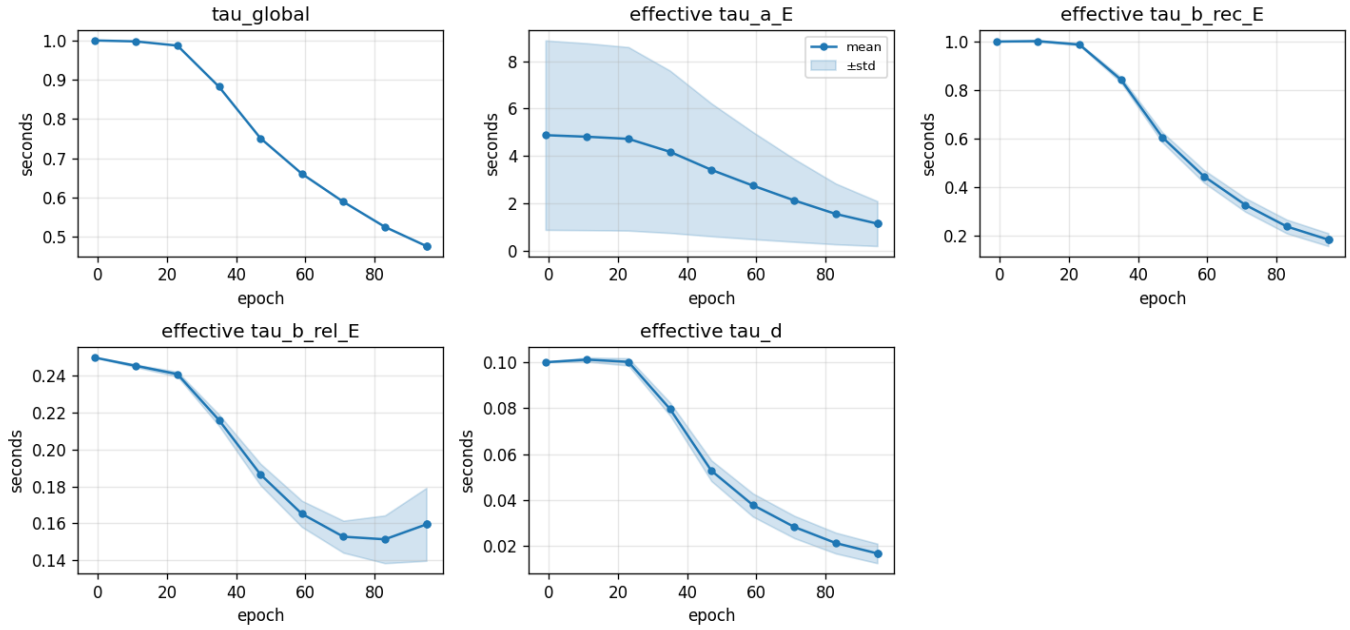
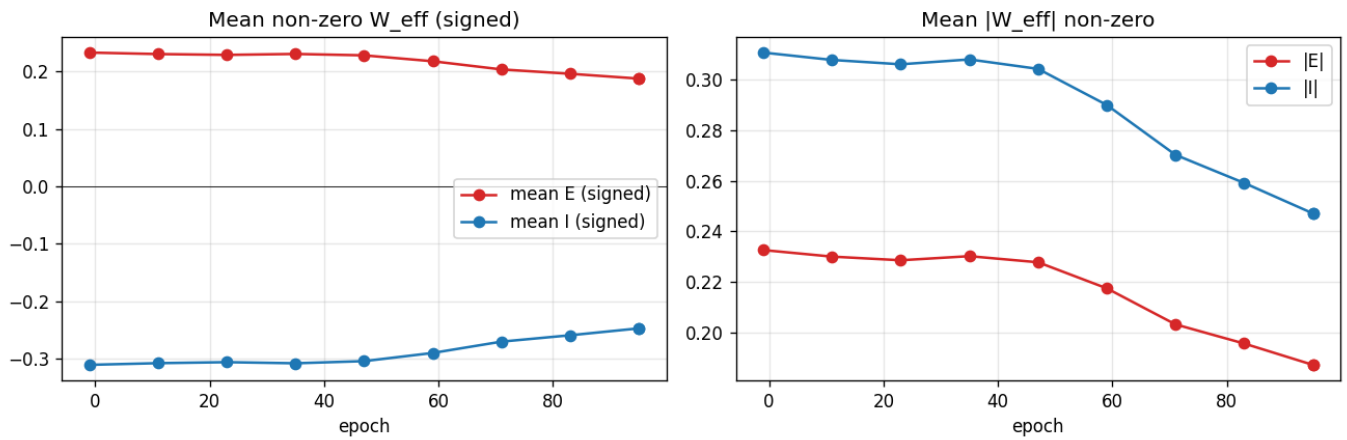


Figure 5: Tau evolution

filt150-whoist-cl — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-e-only-skip-per-neuron (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.47578	-	-	scalar
tau_d (s)	+0.1	+0.016737	0	0.00425	300
tau_a_E[j=0] (s)	+0.25	+0.048089	0	0.01173	150
tau_a_E[j=1] (s)	+4.3832	+1.0284	0	0.01272	150
tau_a_E[j=2] (s)	+10	+2.3605	0	0.003405	150
tau_b_rec_E (s)	+1	+0.18445	0	0.02652	150
tau_b_rel_E (s)	+0.25	+0.15944	0	0.01977	150
W_eff (E src, signed)	+0.23267	+0.18737	0.07641	0.06416	15048
W_eff (I src, signed)	-0.31062	-0.24715	0.07724	0.06573	14977
W_eff (E src)	+0.23267	+0.18737	0.07641	0.06416	15048
W_eff (I src)	+0.31062	+0.24715	0.07724	0.06573	14977
W_in_eff (input neurons)	-0.00084237	-0.00018698	0.1008	0.04121	6675
c_E[j=0] (SFA coupling)	+0.05	+0.087854	0	0.02116	150
c_E[j=1] (SFA coupling)	+0.05	+0.073544	0	0.00723	150
c_E[j=2] (SFA coupling)	+0.05	+0.073719	0	0.007041	150
c_0_E[j=0] (SFA offset)	+0	-0.16547	0	0.119	150
c_0_E[j=1] (SFA offset)	+0	-0.18917	0	0.1404	150
c_0_E[j=2] (SFA offset)	+0	-0.19005	0	0.1373	150
a_0 (threshold)	+0.35	+0.17463	0	0.1116	300
readout_weight	+0.00078219	-0.0060208	0.06654	0.1114	6675
readout_bias	+0	+0.11339	0	0.1413	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

filt150-whoist-cl — effective time constants (srnn-sfa-e-only-per-neuron)

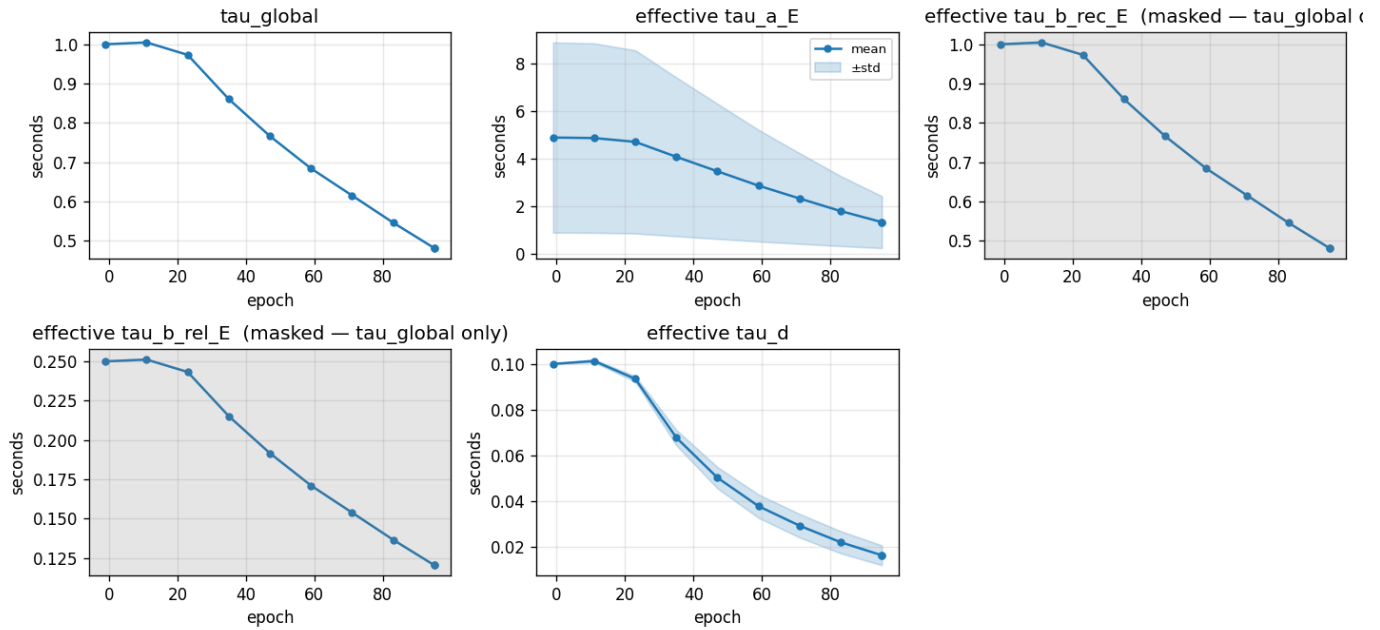


Figure 7: Tau evolution

filt150-whoist-cl — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

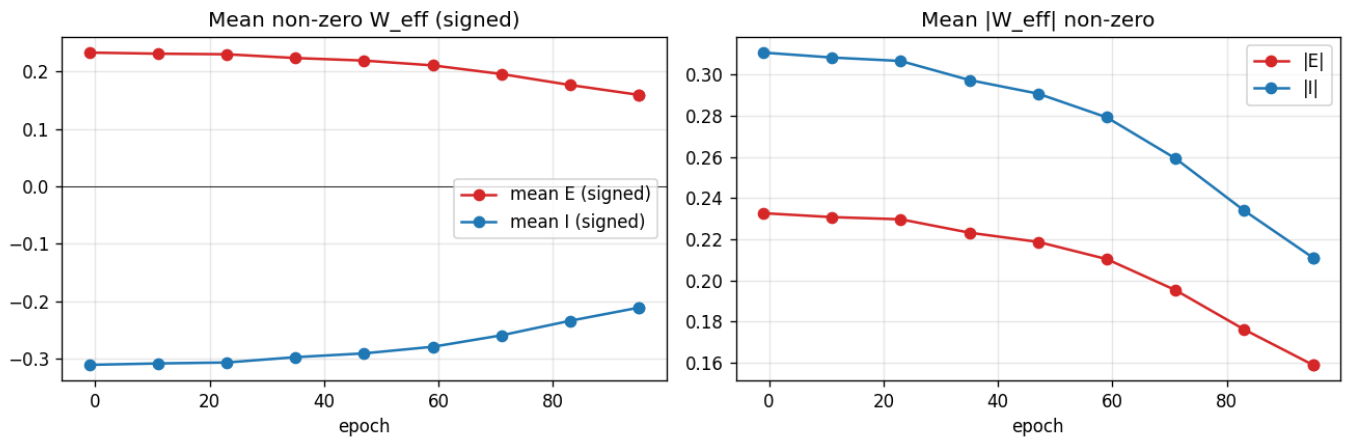


Figure 8: W_EI evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.48155	-	-	scalar
tau_d (s)	+0.1	+0.016454	0	0.004318	300
tau_a_E[j=0] (s)	+0.25	+0.059444	0	0.01066	150
tau_a_E[j=1] (s)	+4.3832	+1.1899	0	0.03563	150
tau_a_E[j=2] (s)	+10	+2.7205	0	0.03012	150
tau_b_rec_E (s)	+1	+0.48155	0	0	150
tau_b_rel_E (s)	+0.25	+0.12039	0	0	150
W_eff (E src, signed)	+0.23267	+0.15906	0.07641	0.05404	15048
W_eff (I src, signed)	-0.31062	-0.21111	0.07724	0.05514	14977
W_eff (E src)	+0.23267	+0.15906	0.07641	0.05404	15048
W_eff (I src)	+0.31062	+0.21111	0.07724	0.05514	14977
W_in_eff (input neurons)	-0.00015382	-0.0018988	0.09934	0.1058	6675
c_E[j=0] (SFA coupling)	+0.05	+0.059055	0	0.01004	150
c_E[j=1] (SFA coupling)	+0.05	+0.055178	0	0.005836	150
c_E[j=2] (SFA coupling)	+0.05	+0.054901	0	0.005454	150
c_0_E[j=0] (SFA offset)	+0	-0.12002	0	0.09616	150
c_0_E[j=1] (SFA offset)	+0	-0.14143	0	0.1268	150
c_0_E[j=2] (SFA offset)	+0	-0.14277	0	0.1289	150
a_0 (threshold)	+0.35	+0.25079	0	0.07603	300
readout_weight	-0.00098972	+0.0080247	0.06678	0.1459	6675
readout_bias	+0	-0.049068	0	0.07967	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

filt150-whoist-cl — effective time constants (srnn-sfa-e-only-skip-per-neuron)

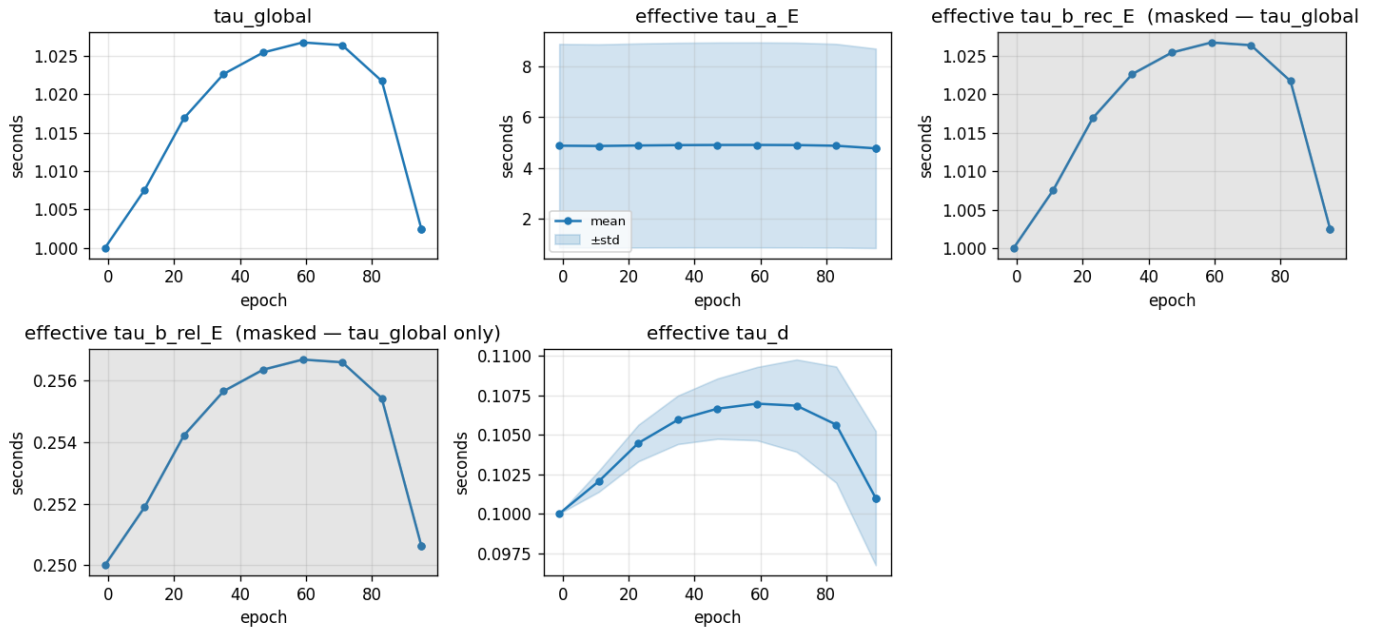
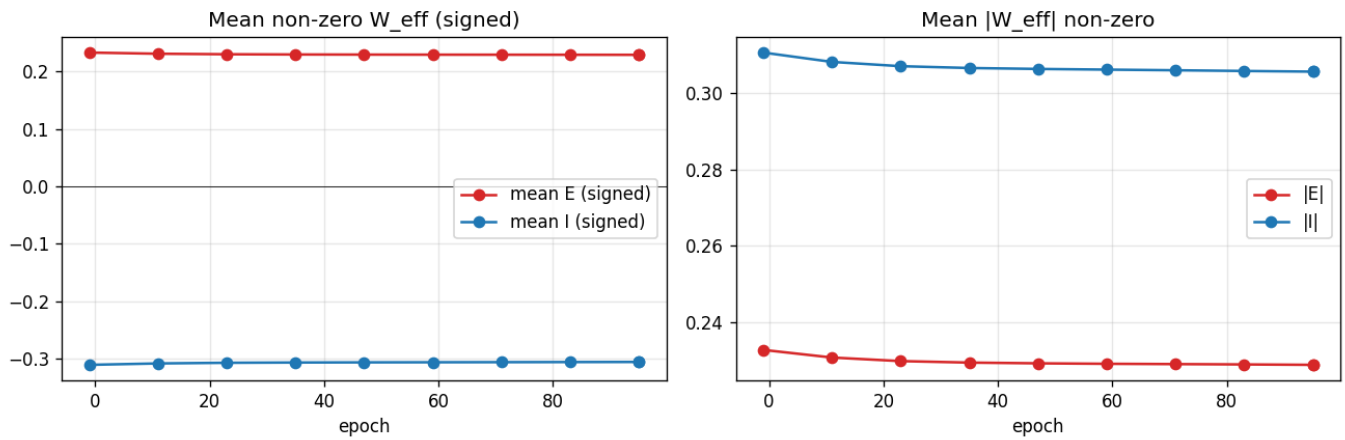


Figure 9: Tau evolution

filt150-whoist-cl — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0025	-	-	scalar
tau_d (s)	+0.1	+0.10099	0	0.004256	300
tau_a_E[j=0] (s)	+0.25	+0.24063	0	0.01187	150
tau_a_E[j=1] (s)	+4.3832	+4.2843	0	0.0225	150
tau_a_E[j=2] (s)	+10	+9.7946	0	0.008763	150
tau_b_rec_E (s)	+1	+1.0025	0	0	150
tau_b_rel_E (s)	+0.25	+0.25062	0	0	150
W_eff (E src, signed)	+0.23267	+0.22876	0.07641	0.07536	15048
W_eff (I src, signed)	-0.31062	-0.30562	0.07724	0.0763	14977
W_eff (E src)	+0.23267	+0.22876	0.07641	0.07536	15048
W_eff (I src)	+0.31062	+0.30562	0.07724	0.0763	14977
W_in_eff (input neurons)	-0.0011621	-0.001351	0.09962	0.09224	6675
c_E[j=0] (SFA coupling)	+0.05	+0.052185	0	0.006185	150
c_E[j=1] (SFA coupling)	+0.05	+0.052166	0	0.002861	150
c_E[j=2] (SFA coupling)	+0.05	+0.052257	0	0.002554	150
c_0_E[j=0] (SFA offset)	+0	+0.041002	0	0.03882	150
c_0_E[j=1] (SFA offset)	+0	+0.043201	0	0.03161	150
c_0_E[j=2] (SFA offset)	+0	+0.043341	0	0.0294	150
a_0 (threshold)	+0.35	+0.38207	0	0.03816	300
readout_weight	+0.00016801	-0.0010836	0.0673	0.06018	6675
readout_bias	+0	+0.0053543	0	0.008919	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

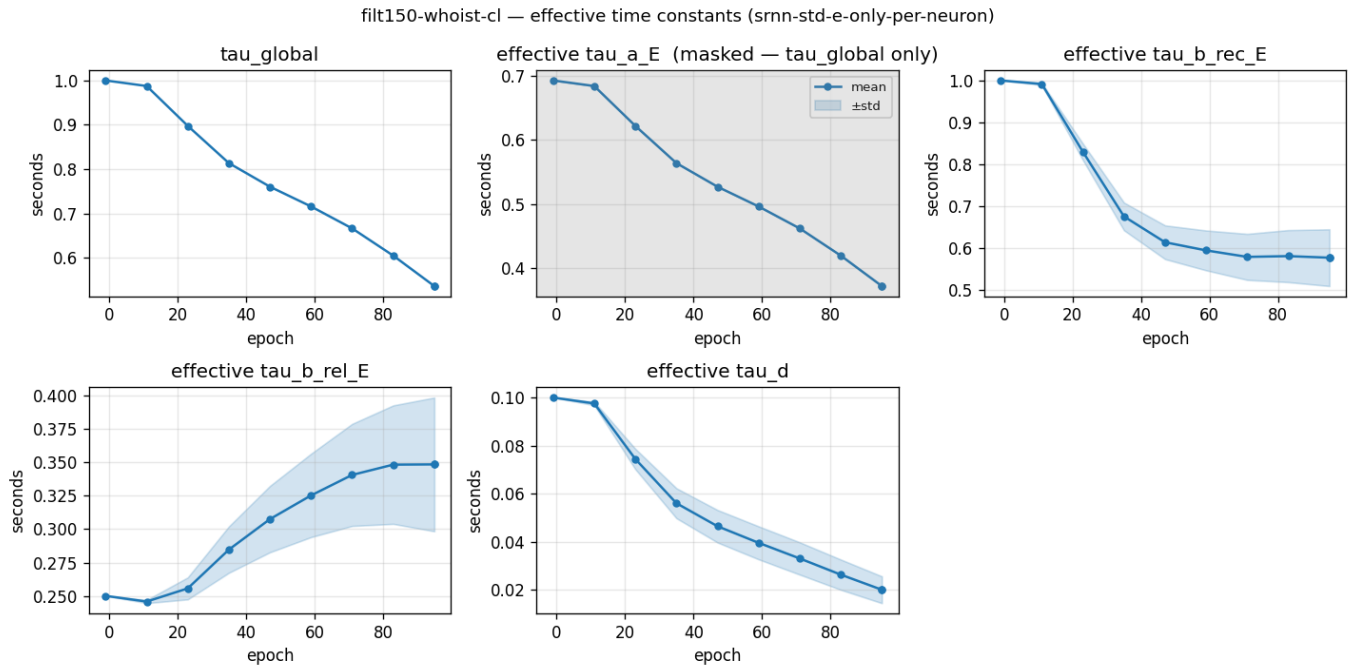


Figure 11: Tau evolution

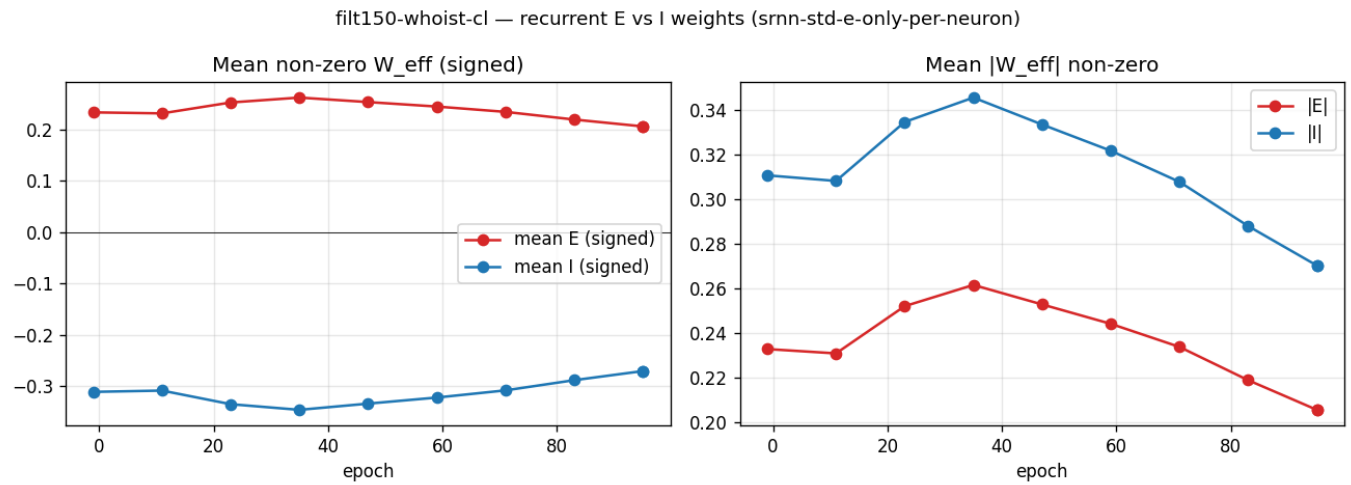


Figure 12: W_EI evolution

srnn-std-e-only-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.53603	-	-	scalar
tau_d (s)	+0.1	+0.020048	0	0.005587	300
tau_b_rec_E (s)	+1	+0.57696	0	0.06774	150
tau_b_rel_E (s)	+0.25	+0.34857	0	0.05015	150
W_eff (E src, signed)	+0.23267	+0.2054	0.07641	0.07192	15048
W_eff (I src, signed)	-0.31062	-0.27018	0.07724	0.07255	14977
W_eff (E src)	+0.23267	+0.2054	0.07641	0.07192	15048
W_eff (I src)	+0.31062	+0.27018	0.07724	0.07255	14977
W_in_eff (input neurons)	+0.00019944	-0.0013044	0.09994	0.1025	6675
a_0 (threshold)	+0.35	-0.11619	0	0.1064	300
readout_weight	+0.00074661	+0.0088996	0.06657	0.1846	6675
readout_bias	+0	-0.064354	0	0.1847	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

filt150-whoist-cl — effective time constants (srnn-std-e-only-skip-per-neuron)

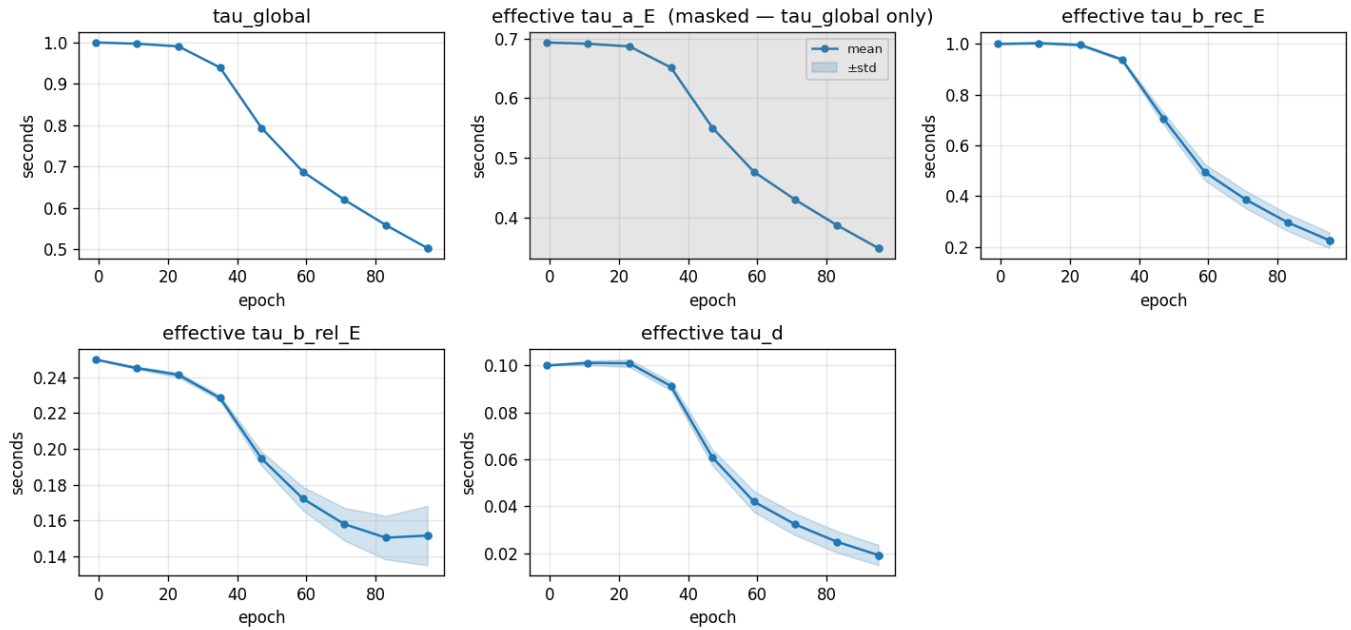


Figure 13: Tau evolution

filt150-whoist-cl — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

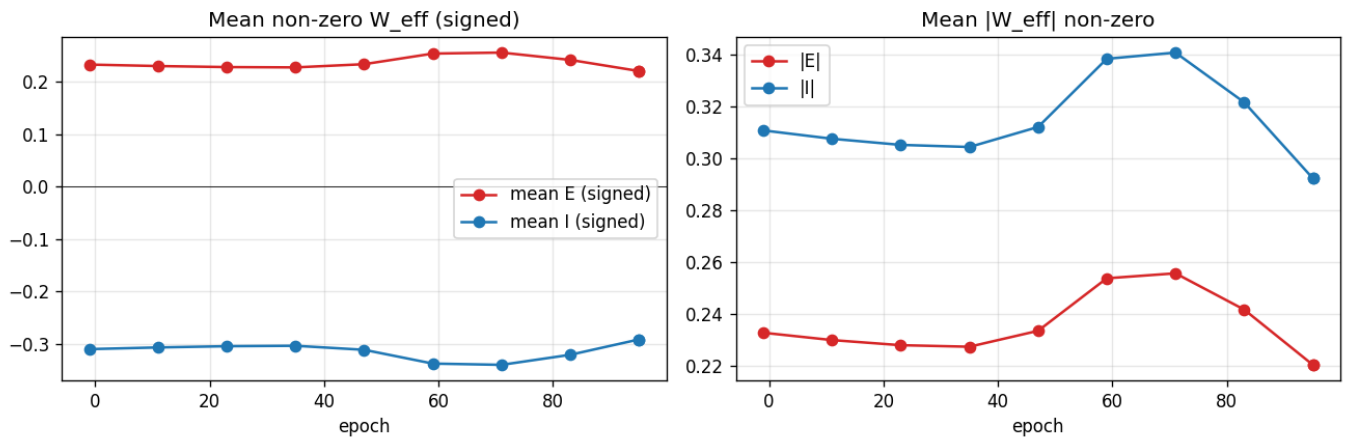


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# filt150-whoist-cl - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.50341	-	-	scalar
tau_d (s)	+0.1	+0.019157	0	0.004312	300
tau_b_rec_E (s)	+1	+0.22586	0	0.03093	150
tau_b_rel_E (s)	+0.25	+0.15146	0	0.01657	150
W_eff (E src, signed)	+0.23267	+0.2203	0.07641	0.07476	15048
W_eff (I src, signed)	-0.31062	-0.29216	0.07724	0.07702	14977
W_eff (E src)	+0.23267	+0.2203	0.07641	0.07476	15048
W_eff (I src)	+0.31062	+0.29216	0.07724	0.07702	14977
W_in_eff (input neurons)	+0.00036119	+0.00040633	0.09956	0.04801	6675
a_0 (threshold)	+0.35	+0.25135	0	0.1014	300
readout_weight	+0.00055372	+0.00015984	0.06572	0.09561	6675
readout_bias	+0	-0.028081	0	0.1108	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900